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SUBJECT CODE NO: - RNP-8008
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B]Mech, Civil, Comp. AI & ML,EEE
Examination January/February 2025
Design Thinking

[Time: 2:00 Hours]**[Max. Marks:30]**

Please check whether you have got the right question paper.

N. B

- 1) Question No 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

Part A**Q1 Attempt any Five****10**

- i. Why is the "Empathize" stage crucial in the Design Thinking process?
 - a) To identify potential roadblocks
 - b) To analyze competitor products
 - c) To understand the needs and emotions of users
 - d) To determine the project budget
- ii. What distinguishes Design Thinking from traditional problem-solving methods?
 - a) Rigid adherence to predefined steps
 - b) Focus on tangible results only
 - c) Continuous iteration and learning from failures
 - d) Limited involvement of end-users
- iii. Design Thinking is most effective when applied to problems that are:
 - a) Simple and straightforward
 - b) Ill-defined or complex
 - c) Purely technical
 - d) Well-documented with clear solutions
- iv. Empathy maps are divided into which categories?
 - a) Technical, Operational, Financial, Emotional
 - b) Thinking, Doing, Observing, Reacting
 - c) Says, Thinks, Feels, Does
 - d) Needs, Wants, Actions, Reactions State true or false
- v. During which stage would you: Create a model of your solution.

a) Prototype	b) Define
c) Ideate	d) Test
- vi. Which of the following is an example of a low-fidelity prototype?
 - a) A fully functional mobile app
 - b) A rough paper sketch
 - c) A digital wireframe with animations
 - d) A complete 3D-printed model

- vii. The question "How might we...?" is often used to craft _____ problem statements.
- viii. _____ is a group activity where participants generate ideas without judgment.

Part B

Solve any four from the following

- Q2** What is empathy in Design Thinking, and why is it important for user-centered design? **5**
- Q3** How Is Design Thinking Different From Traditional Think **5**
- Q4** Explain Empathy mapping **5**
- Q5** Explain the process of defining a problem in Design Thinking and its importance. **5**
- Q6** What is prototyping, and why is it an essential part of the Design Thinking process? **5**
- Q7** Why is testing considered a critical phase in iterative and user-centered design? **5**